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Project Report

Emotion Classification in Text

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1. Introduction

Emotion classification in natural language text has numerous practical applications, including opinion mining, public sentiment monitoring, and personalized recommendation systems. This task becomes particularly challenging when texts are short and informal. The objective of this project is to evaluate and compare the effectiveness of different text classification approaches on a Twitter-derived benchmark dataset [1], which accurately reflects these challenges.

Input text	**Assigned emotion**
when i heard a rumour that the st year exam results were out i had fear that i might be one of the failures	fear
i already have my christmas trees up i got two and am feeling festive which i m sure is spurring me to get started on this book	joy
i go online and i see a friend talking to another one and is not talking to me i feel ignored i feel unloved	sadness

Table 1.1. Representative samples from the dataset.

In this work, three distinct classification paradigms were implemented, trained, and evaluated:

1. **Classical feature extraction + Multi-Layer Perceptron (MLP):** Text representations extracted via TF-IDF followed by a feed-forward neural network classifier.
2. **Encoder-only pre-trained Transformer:** Probabilistic sequence classification based on fine-tuning `distilbert-base-uncased`.
3. **Decoder-only autoregressive LLM:** Adapting `Qwen3-1.7B-Base` to generate categorical label predictions using few-shot prompting and Low-Rank Adaptation (LoRA).

For each approach, standard classification metrics—accuracy, precision, recall, and F1-score (per-class, macro, and weighted averages)—as well as confusion matrices were computed on both the training and test splits. Furthermore, low-dimensional feature space representations were extracted and visualized for the test set using UMAP.

2. Related Work

In Approach 2, the `distilbert-base-uncased` model [2] was utilized, whereas Approach 3 leverages the `Qwen3-1.7B-Base` model [3].

The dataset was originally curated for the CARER (*Contextualized Affect Representations for Emotion Recognition*) framework [1]. The authors introduced a graph-based method capable of extracting rich structural text representations that capture affect while preserving context. The dataset consists of publicly posted Twitter tweets that were automatically annotated with emotion labels based on associated emotional hashtags.

3. Methodology

3.1. Tools and Libraries

Category	Libraries and Tools
Visualization	`umap-learn`, `matplotlib`, `seaborn`
Dataset Management	`datasets`
Modeling, Training, & Adaptation	`torch`, `lightning`, `transformers`, `peft`, `bitsandbytes`, `accelerate`
Evaluation Metrics	`torchmetrics`, `evaluate`, `scikit-learn`
Utilities	`tqdm`, `numpy`, `pandas`, `colorama`, `wandb`, `scipy`

Table 3.1. Overview of software libraries and frameworks used in the project.

3.2. Dataset Description

The dataset is accessible via the Hugging Face Hub at `https://huggingface.co/datasets/dair-ai/emotion`. It contains two primary columns: `text` (the tweet content in English) and `label` (the categorical emotion index from 0 to 5). The target classes are defined as:

- 0 – *sadness*,
- 1 – *joy*,
- 2 – *love*,
- 3 – *anger*,
- 4 – *fear*,
- 5 – *surprise*.

The standard split provided by the benchmark was retained: 16,000 samples for training, 2,000 samples for validation, and 2,000 samples for testing. The class distribution across all splits is shown in Figure 3.1.

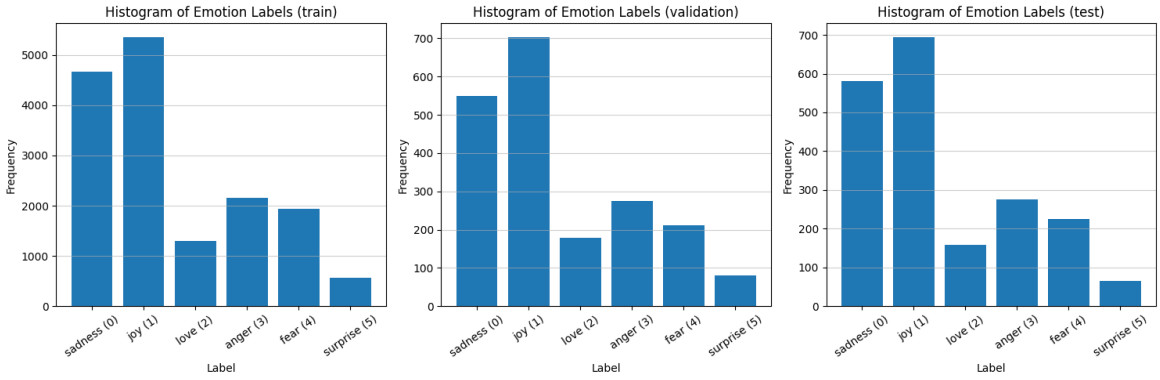


Figure 3.1. Class distribution across training, validation, and test splits.

3.3. Approach 1: TF-IDF with Multi-Layer Perceptron (MLP)

Feature extraction was performed using `TfidfVectorizer` from `scikit-learn` with the following hyperparameters:

- `max_features=10000`
- `lowercase=True`
- `analyzer='word'`
- `ngram_range=(1,2)`
- `stop_words='english'`

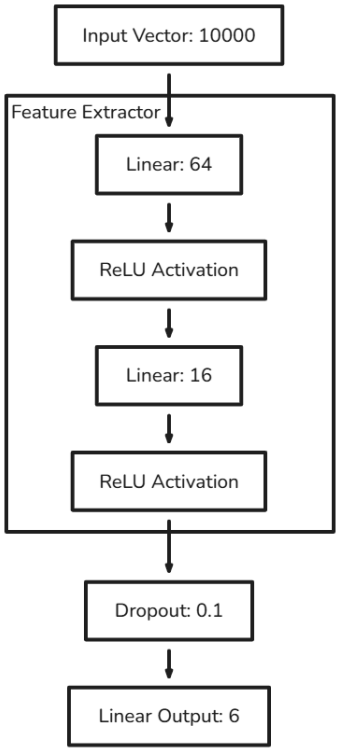


Figure 3.2. Architecture diagram of the EmotionNet MLP model.

The vocabulary and IDF weights were fitted strictly on the training partition and subsequently applied to transform the validation and test sets. Transformed sparse vectors were wrapped in PyTorch `DataLoader` objects. The neural network classifier (`EmotionNet`) was constructed with PyTorch using dense linear layers, non-linear activations, and batch normalization/dropout, as detailed in Figure 3.2.

The training process was orchestrated using PyTorch Lightning with the `AdamW` optimizer and a `CosineAnnealingLR` learning rate scheduler.

3.3.1. EmotionNet Training Dynamics

Training dynamics were monitored using Cross-Entropy loss and classification accuracy.



Figure 3.3. Training Loss

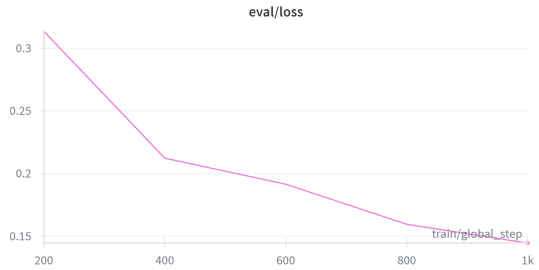


Figure 3.4. Validation Loss

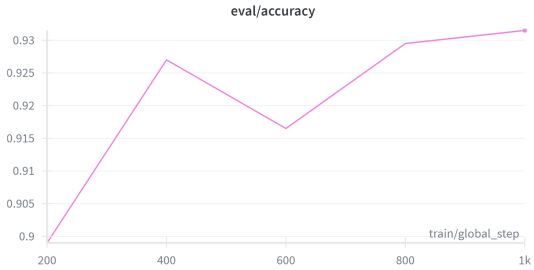


Figure 3.5. Validation Accuracy

Figure 3.6. Training curves for the EmotionNet MLP model.

3.4. Approach 2: Fine-Tuned DistilBERT Classifier

In this approach, the pre-trained `distilbert-base-uncased` model (`https://huggingface.co/distilbert/distilbert-base-uncased`) was instantiated using `AutoModelForSe` attaching a sequence classification head initialized on top of the pooled Transformer representations. Fine-tuning was executed with the Hugging Face `Trainer` API.

3.4.1. DistilBERT Training Dynamics

The model was fine-tuned while tracking validation loss and classification accuracy across training checkpoints.

Table 3.2. Fine-tuning progress for the DistilBERT model.

Step	Training Loss	Validation Loss	Accuracy
200	0.7841	0.313556	0.8990
400	0.2555	0.212283	0.9270
600	0.1733	0.191464	0.9165
800	0.1250	0.159521	0.9295
1000	0.1268	0.144575	0.9315

3.5. Approach 3: Adapting the Autoregressive Qwen3-1.7B-Base Model

For the generative LLM approach, `Qwen3-1.7B-Base` (`https://huggingface.co/Qwen/Qwen3-1.7B-Base`) was loaded using `AutoModelForCausalLM`. Two adaptation strategies were examined:

- Few-shot prompting
- Low-Rank Adaptation (LoRA) parameter-efficient fine-tuning

The prompt template designed for inference was structured as follows:

```
Your task is to analyze emotions in text.
Assign one of these 6 emotions: ['sadness', 'joy', 'love', 'anger',
'fear', 'surprise'] to the following text.
Answer only with one word from emotions list.
## Input Text: <input text from dataset>
## Assigned emotion:
```

For supervised fine-tuning, training instances were formatted with the target completion followed by an end-of-sequence delimiter:

```
Your task is to analyze emotions in text.
Assign one of these 6 emotions: ['sadness', 'joy', 'love', 'anger',
'fear', 'surprise'] to the following text.
Answer only with one word from emotions list.
## Input Text: <input text from dataset>
## Assigned emotion: <ground truth label> <|endoftext|>
```

Under the few-shot prompting setup, each prompt provided 3 demonstrations sampled from the training partition followed by 1 query sample from the test set.

For LoRA fine-tuning, adapter rank $r = 8$ and $\alpha = 32$ were configured. Adapters were injected into all attention projection and feed-forward linear layers: `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`. Training was conducted via the Hugging Face `Trainer` with causal language modeling objectives.

3.5.1. Qwen3 LoRA Training Dynamics

Training progress was monitored using Cross-Entropy loss over generated target tokens.



Figure 3.7. Training Loss

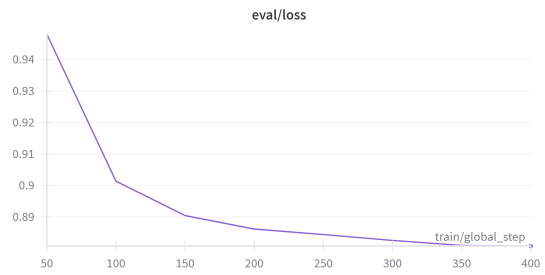


Figure 3.8. Validation Loss

Figure 3.9. Loss trajectories during LoRA fine-tuning of Qwen3-1.7B-Base.

4. Experimental Evaluation and Results

To evaluate model performance across configurations, the following metrics were recorded:

- Per-class Precision, Recall, and F1-score
- Overall Accuracy
- Macro-averaged Precision, Recall, and F1-score
- Weighted-averaged Precision, Recall, and F1-score
- Confusion Matrices

All models were evaluated on both the training and test sets. For few-shot prompting with Qwen3, exemplars were drawn from the training partition and evaluation was carried out directly on the test set. Additionally, formatting compliance was tracked: generations failing to produce one of the six valid class labels were filtered out. Under few-shot prompting, 3 test samples produced invalid formats; under LoRA fine-tuning, 1 test sample and 6 training samples were discarded due to invalid outputs.

Finally, 2D latent representations were extracted from intermediate hidden states and visualized using UMAP on the test split.

4.1. Performance Metrics

Table 4.1. Per-class F1-Score comparison across models and splits.

Model and Split	sadness	joy	love	anger	fear	surprise
TF-IDF MLP (test)	0.91	0.91	0.68	0.86	0.86	0.11
DistilBERT (test)	**0.97**	**0.96**	**0.86**	**0.92**	**0.89**	**0.75**
Qwen Few-Shot (test)	0.50	0.68	0.29	0.48	0.36	0.16
Qwen LoRA (test)	0.68	0.76	0.30	0.56	0.55	0.00
DistilBERT (train)	**0.99**	0.97	**0.90**	0.97	0.94	**0.87**
Qwen LoRA (train)	0.69	0.73	0.30	0.61	0.57	0.08
TF-IDF MLP (train)	**0.99**	**0.99**	0.84	**0.98**	**0.97**	0.18

Table 4.2. Per-class Precision comparison across models and splits.

Model and Split	sadness	joy	love	anger	fear	surprise
TF-IDF MLP (test)	0.88	0.89	0.66	0.89	0.85	**1.00**
DistilBERT (test)	**0.96**	**0.95**	**0.87**	**0.94**	**0.87**	0.75
Qwen Few-Shot (test)	0.73	0.69	0.38	0.47	0.35	0.09
Qwen LoRA (test)	0.61	0.72	0.47	0.66	0.60	0.00
DistilBERT (train)	0.98	0.97	**0.91**	**0.97**	0.94	0.85
Qwen LoRA (train)	0.61	0.68	0.48	0.69	0.64	0.96
TF-IDF MLP (train)	**0.99**	**0.99**	0.74	**0.97**	**0.96**	**1.00**

Table 4.3. Per-class Recall comparison across models and splits.

Model and Split	sadness	joy	love	anger	fear	surprise
TF-IDF MLP (test)	0.94	0.92	0.70	0.84	0.86	0.06
DistilBERT (test)	**0.97**	**0.96**	**0.84**	**0.90**	**0.90**	**0.76**
Qwen Few-Shot (test)	0.38	0.68	0.23	0.49	0.36	0.56
Qwen LoRA (test)	0.78	0.81	0.22	0.49	0.51	0.00
DistilBERT (train)	**0.99**	0.97	0.89	0.96	0.95	**0.89**
Qwen LoRA (train)	0.78	0.78	0.22	0.54	0.52	0.04
TF-IDF MLP (train)	**0.99**	**0.99**	**0.97**	**0.98**	**0.98**	0.10

Table 4.4. Comprehensive summary of aggregated performance metrics (P: Precision, R: Recall).

Model and Split	Accuracy	Macro P	Macro R	Macro F1	Weighted P	Weighted R	Weighted F1
TF-IDF MLP (test)	0.86	0.86	0.72	0.72	0.87	0.86	0.85
DistilBERT (test)	**0.93**	**0.89**	**0.89**	**0.89**	**0.93**	**0.93**	**0.93**
Qwen Few-Shot (test)	0.49	0.45	0.45	0.41	0.59	0.49	0.52
Qwen LoRA (test)	0.65	0.51	0.47	0.48	0.62	0.65	0.63
TF-IDF MLP (train)	0.95	**0.94**	0.83	0.82	**0.96**	0.95	0.94
Qwen LoRA (train)	0.65	0.68	0.48	0.50	0.65	0.65	0.62
DistilBERT (train)	**0.96**	**0.94**	**0.94**	**0.94**	**0.96**	**0.96**	**0.96**

4.2. Confusion Matrices

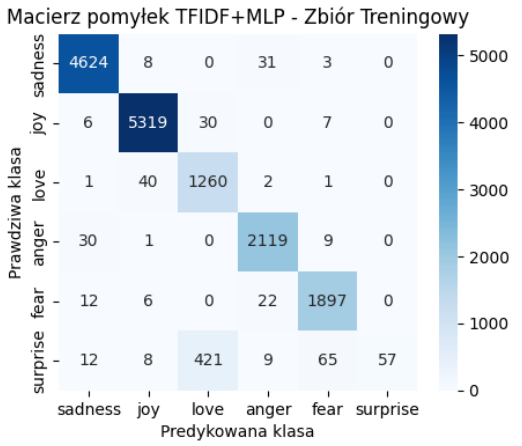


Figure 4.1. Confusion Matrix: TF-IDF + MLP (Train)

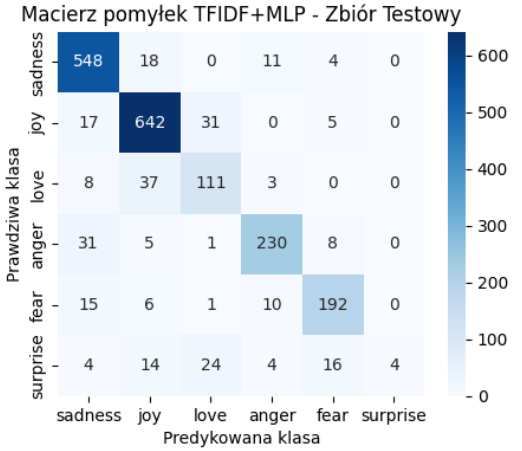


Figure 4.2. Confusion Matrix: TF-IDF + MLP (Test)

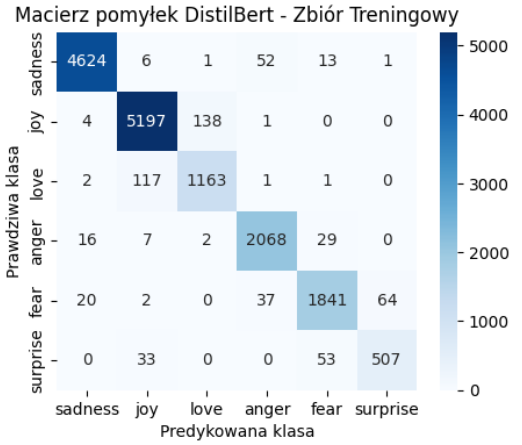


Figure 4.3. Confusion Matrix: DistilBERT (Train)

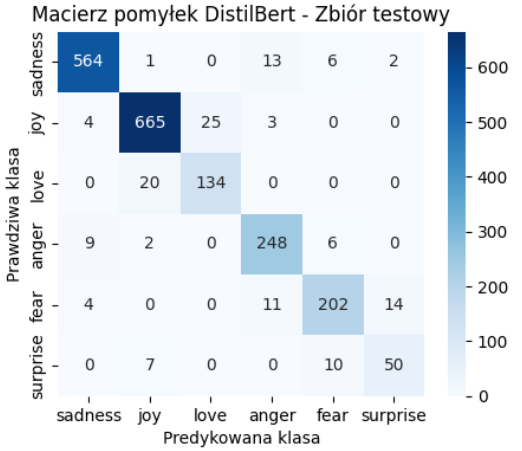


Figure 4.4. Confusion Matrix: DistilBERT (Test)

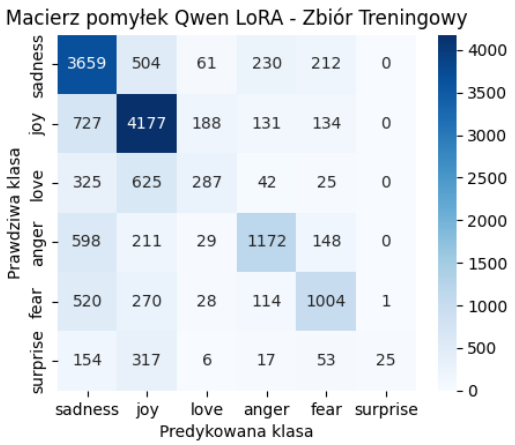


Figure 4.5. Confusion Matrix: Qwen3 LoRA (Train)

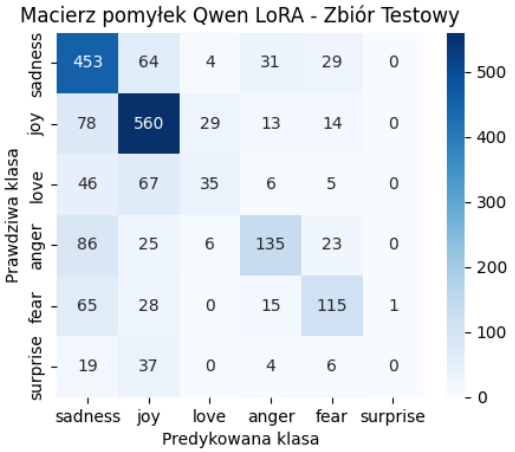


Figure 4.6. Confusion Matrix: Qwen3 LoRA (Test)

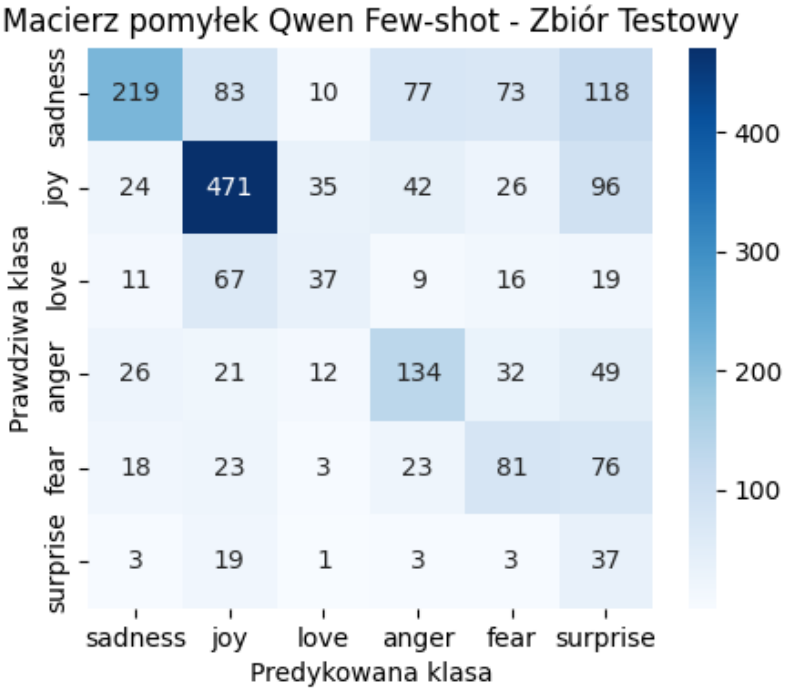


Figure 4.7. Confusion Matrix: Qwen3 Few-Shot Prompting (Test)

4.3. UMAP Representation Visualizations

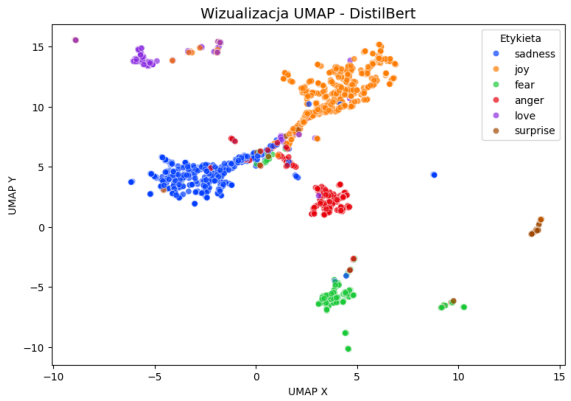


Figure 4.8. UMAP Representation: DistilBERT

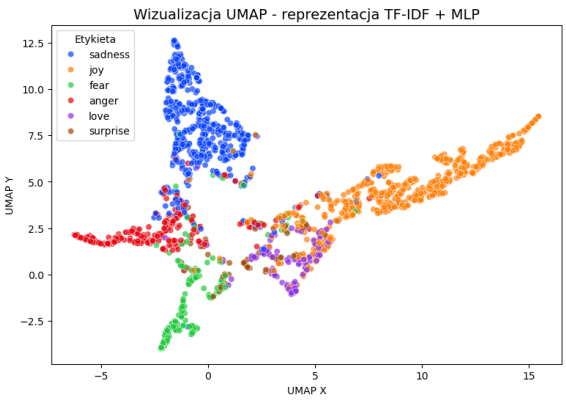


Figure 4.9. UMAP Representation: TF-IDF + MLP

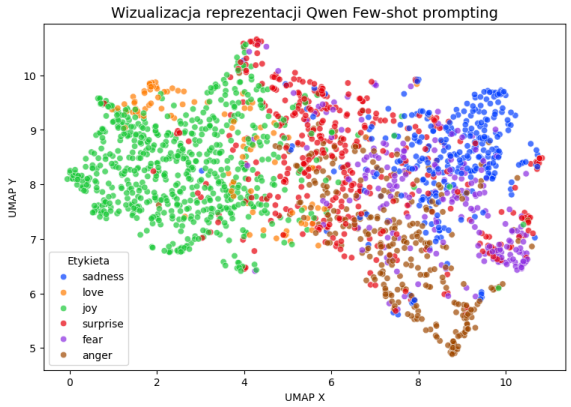


Figure 4.10. UMAP Representation: Qwen3 Few-Shot

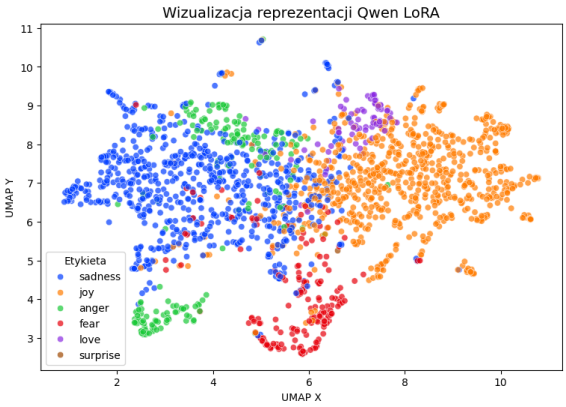


Figure 4.11. UMAP Representation: Qwen3 LoRA

4.4. Discussion of Results

The highest performance among all tested solutions was achieved by the DistilBERT model. It obtained the highest accuracy of 0.93 for the test set and 0.96 for the training set. The MLP model based on TF-IDF representations, despite relying on a simpler text representation, achieved a very solid result on the test set (Accuracy 0.86). This is significantly better than both variants of the Qwen model, showing that classical text classification methods can still be very competitive. The use of a decoder-only model (Qwen3-1.7B) yielded the worst results of all approaches.

- Few-shot prompting proved to be the weakest method (Accuracy 0.49). This model had particular difficulty distinguishing classes other than joy, as seen in the UMAP visualization – the labels surprise, fear, sadness, and anger are heavily mixed together.
- Fine-tuning with LoRA improved results compared to prompting (Accuracy increased to 0.65).

The under-represented ***surprise*** class presented the steepest challenge across all paradigms:

- DistilBERT handled class imbalance most effectively, maintaining an F1-score of 0.75 on the test set.
- While TF-IDF + MLP attained 1.00 precision on *surprise*, its recall dropped to 0.06, indicating that the network only predicted this class under extreme certainty.
- Qwen3 fine-tuned with LoRA collapsed entirely on *surprise* on the test partition (F1-score 0.00), demonstrating vulnerability to severe class imbalance during generative alignment.

Dominant classes (*sadness* and *joy*) consistently exhibited the highest recognition accuracy across all architectures due to their strong representation in the training distribution.

5. Summary

The conducted research on emotion classification in text allowed for the comparison of three different approaches: a classical method based on vector representation (TF-IDF with an MLP network), an encoder-only model (DistilBERT), and a decoder-only model (Qwen3-1.7B).

The short and informal nature of the classified texts did not pose a problem for the first two approaches, so the stated problem was successfully solved. The fine-tuned DistilBERT model achieved the highest performance. This model was able to recognize even less frequent classes, such as surprise. The approach based on TF-IDF and MLP also achieves high results, but performs worse on classes that occur less frequently in the dataset. The selected adaptation methods for the Qwen3 model proved to be ineffective; the model achieved the worst results among the 3 compared approaches.

As part of further development of the project, it is worth considering the use of larger models from the Qwen family or applying other fine-tuning techniques.

References

[1] E. Saravia, H.-C. T. Liu, Y.-H. Huang, J. Wu, and Y.-S. Chen, "Carer: Contextualized affect representations for emotion recognition", in *Proceedings of the 2018 conference on empirical methods in natural language processing*, 2018, pp. 3687–3697.

[2] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "Distilbert, a distilled version of bert: Smaller, faster, cheaper and lighter", *arXiv preprint arXiv:1910.01108*, 2019.

[3] A. Yang et al., "Qwen3 technical report", *arXiv preprint arXiv:2505.09388*, 2025.